



NORTH SYDNEY BOYS HIGH SCHOOL

2010 HSC ASSESSMENT TASK 2

Mathematics Extension 2

General Instructions

- Working time – **60 minutes**
- Write in the booklet provided on both sides of the paper
- Write using blue or black pen
- Board approved calculators may be used
- All necessary working should be shown in every question
- Each new question is to be started on a **new page**.

- Attempt all questions

Class Teacher:

(Please tick or highlight)

- ☐ Mr Barrett
☐ Mr Trenwith
☐ Mr Weiss

Student Number: _____

(To be used by the exam markers only.)

Question No	1	2	3	4	5	6	7	Total	Total %
Mark	$\frac{5}{5}$	$\frac{15}{15}$	$\frac{4}{4}$	$\frac{5}{5}$	$\frac{6}{6}$	$\frac{4}{4}$	$\frac{6}{6}$	$\frac{45}{45}$	$\frac{100}{100}$

Question 1 (5 marks)

Sketch a graph of $\frac{x^2}{9} - \frac{y^2}{16} = 1$, showing the coordinates of the vertices and foci, and the equations of the directrices and asymptotes. 5

Question 2 (15 marks)

- (a) Sketch a graph of $x = 2t$, $y = \frac{2}{t}$, showing the coordinates of the vertices and foci, and the equations of the directrices. 3
- (b) Show that the equation of the tangent to this curve at the point P where $t = p$ is $x + p^2y = 4p$. 2
- (c) This tangent intersects the x -axis at A , and the y -axis at B , and O is the origin. Show that the area of triangle OAB is independent of the position of P . 3
- (d) Q is the point on the curve where $t = q$. 3
Show that the tangents at P and Q intersect at $T\left(\frac{4pq}{p+q}, \frac{4}{p+q}\right)$.
- (e) The chord PQ has a gradient of $-\frac{1}{2}$. 2
Show that $pq = 2$.
- (f) Hence find the Cartesian equation for the locus of T . 2

Question 3 (4 marks)

- (a) Write down **in integral form** an expression for the area enclosed by the ellipse $\frac{x^2}{9} + \frac{y^2}{4} = 1$. 2
- (b) **Hence**, find the area enclosed by the ellipse. 2

Question 4 (5 marks)

$P(a \cos \theta, b \sin \theta)$ is a point on the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.

- (a) Show that the equation of the normal to P is: 3

$$ax \sin \theta - by \cos \theta = (a^2 - b^2) \sin \theta \cos \theta$$

- (b) G is the point where this normal meets the x -axis, N is the foot of the perpendicular from P to the x -axis, and O is the origin. 2

Show that $\frac{OG}{ON} = e^2$.

Question 5 (6 marks)

- (a) Decompose $\frac{x^2 - x + 2}{(x + 1)(x^2 + 1)}$ into partial fractions. 3

- (b) Hence find $\int \frac{x^2 - x + 2}{(x + 1)(x^2 + 1)} dx$, writing your answer in the form $\ln[f(x)] + c$. 3

Question 6 (4 marks)

It can be shown that $1^2 + 2^2 + 3^2 + \dots + n^2 = \frac{1}{6}n(n + 1)(2n + 1)$.

[DO NOT prove this result.]

Prove by mathematical induction (using the result above) that:

$$1^2 + (1^2 + 2^2) + (1^2 + 2^2 + 3^2) + \dots + \sum_{k=1}^n k^2 = \frac{n}{12}(n + 1)^2(n + 2)$$

Question 7 (6 marks)

- (a) Show that if $y = mx + k$ is a tangent to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, then $a^2m^2 - b^2 = k^2$. 3

- (b) Hence find the equation of the tangents from the point $(1, 3)$ to the hyperbola $\frac{x^2}{4} - \frac{y^2}{15} = 1$, and the coordinates of their points of contact. 3

Question 1

5

$$\frac{x^2}{9} - \frac{y^2}{16} = 1$$

$$ae = 3 \times \frac{5}{3} = 5$$

$$b^2 = a^2(e^2 - 1)$$

$$16 = 9(e^2 - 1)$$

$$e^2 - 1 = \frac{16}{9}$$

$$e^2 = \frac{25}{9}$$

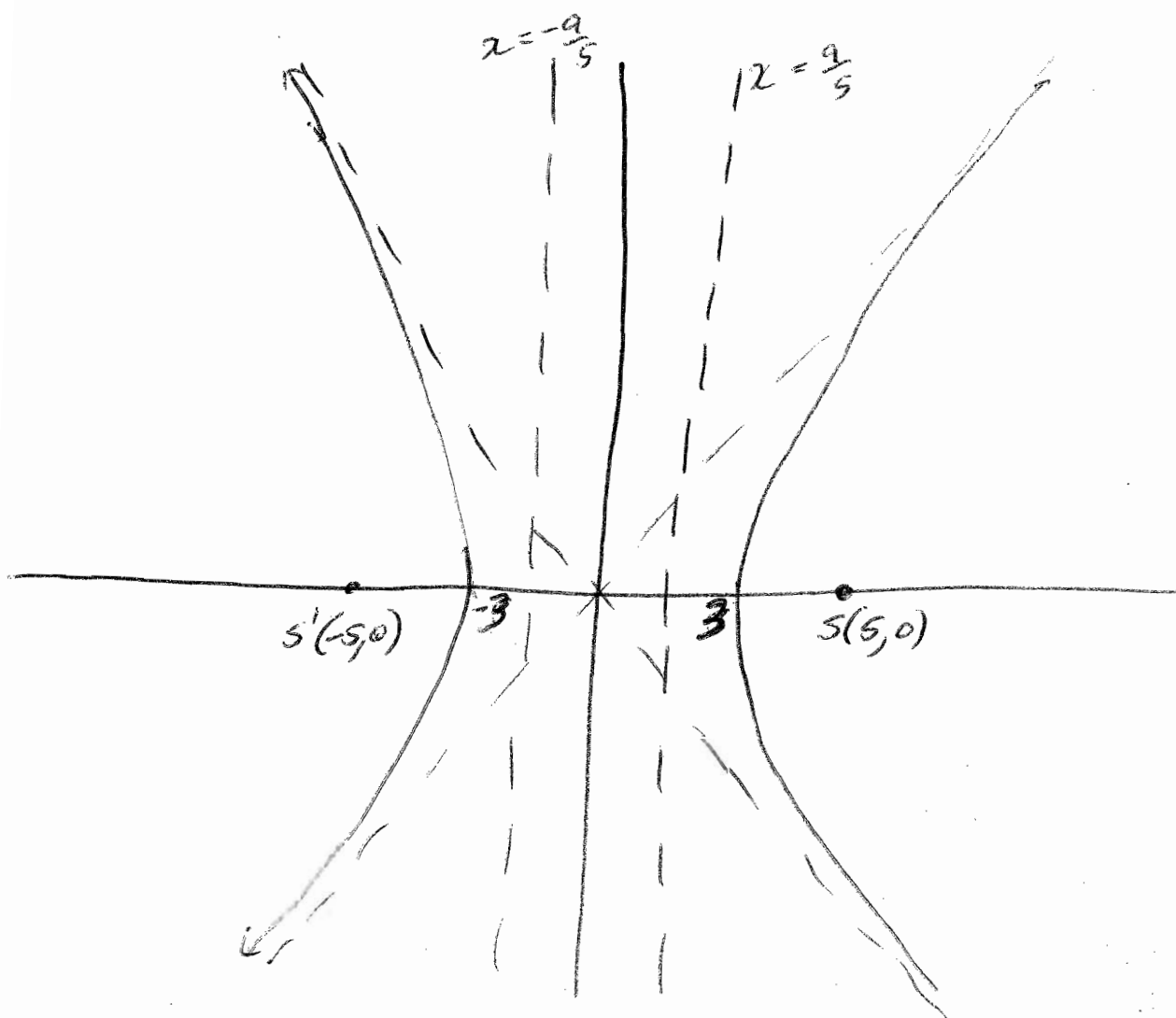
$$e = \frac{5}{3} - [1]$$

$$\therefore \text{Foci } (\pm 5, 0) - [1]$$

$$\frac{a}{e} = 3 \times \frac{3}{5} = \frac{9}{5}$$

$$\text{directrices : } x = \pm \frac{9}{5} - [1]$$

$$\text{asymptotes : } y = \pm \frac{4}{3}x - [1]$$

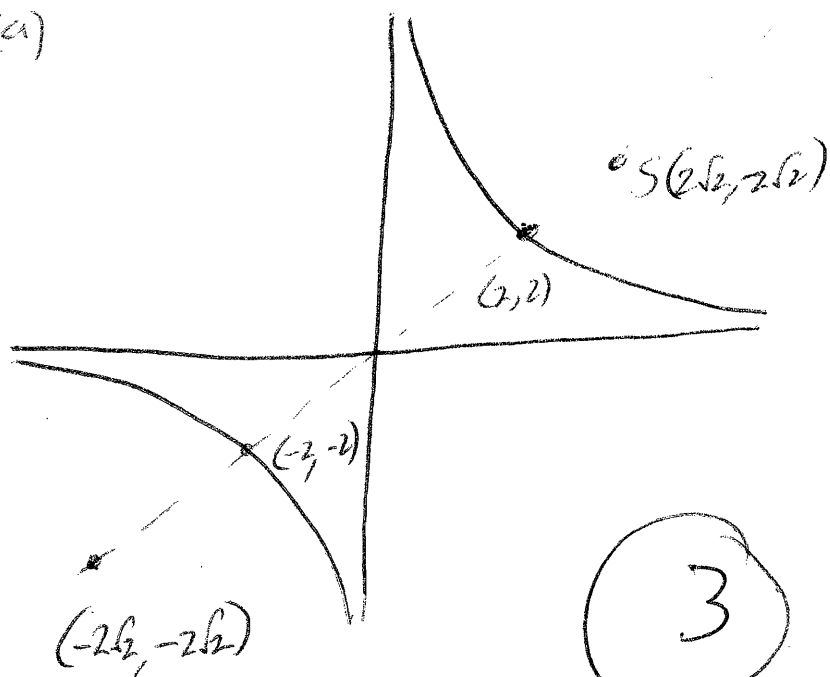


- [1] for correct shape & correct vertices (don't penalise if they write an x-int of 3 instead of $(3, 0)$)

Question 2

15

(a)



$$xy = 4$$

$$V(\pm 2, \pm 2) \quad [1]$$

$$S(\pm 2\sqrt{2}, \pm 2\sqrt{2}) \quad [1]$$

$$\text{asymptotes: } x+y = \pm 2\sqrt{2}$$

[1]

3

$$(b) \quad x = 2t \quad y = \frac{2}{t}$$

$$\frac{dx}{dt} = 2$$

$$\frac{dy}{dt} = -\frac{2}{t^2}$$

$$\frac{dy}{dx} = \frac{-\frac{2}{t^2}}{2} \cdot \frac{1}{2}$$

$$= -\frac{1}{t^2}$$

$$\therefore m_T = -\frac{1}{p^2} \quad [1]$$

$$y - \frac{2}{p} = -\frac{1}{p^2}(x - 2p)$$

$$p^2y - 2p = -x + 2p$$

$$x + p^2y = 4p \quad [1]$$

2

$$(c) \quad x + y = 4p \quad \Rightarrow \quad x = 4p \quad - [1]$$

$$y + x = 4q \quad \Rightarrow \quad y = \frac{4}{p} \quad - [1]$$

$$\text{Area} = \frac{1}{2} \times 4p \times \frac{4}{p} \\ = 8 \text{ units}^2 \quad (\text{which is indep of } p) \quad [1]$$

$$(d) \quad x + y = 4p$$

$$x + q^2 y = 4q$$

$$(e) \quad (p - q^2)y = 4(p - q) \quad [1]$$

$$y = \frac{4}{p + q}$$

[1]

$$\rightarrow x + \frac{4p^2}{p + q} = 4p$$

$$x = \frac{4p(p + q) - 4p^2}{p + q}$$

$$x = \frac{4pq}{p + q} \quad [1]$$

$$[1] \quad \therefore T \left[\frac{4pq}{p + q}, \frac{4}{p + q} \right]$$

$$(e) \quad m_{pq} = -\frac{1}{2}$$

$$\frac{\frac{2}{p} - \frac{2}{q}}{2p - 2q} = -\frac{1}{2} \quad [1]$$

cross-multiply: $\frac{4}{p} - \frac{4}{q} = 2q - 2p$

$$\times pq: \quad 4q - 4p = 2pq(q - p)$$

$$4(q - p) = 2pq(q - p)$$

$$pq = 2$$

[1]

$$(f) \quad pq = 2 \Rightarrow x = \frac{8}{p+q} \quad / \quad y = \frac{4}{p+q}$$

$$pq = \frac{8}{x} \Rightarrow y = 4 \times \frac{x}{8}$$

[1]

$$y = \frac{x}{2}$$

[1]

Question 3

$$(a) \quad \frac{x^2}{9} + \frac{y^2}{4} = 1$$

$$\frac{y^2}{4} = 1 - \frac{x^2}{9}$$

$$= \frac{9 - x^2}{9}$$

$$y^2 = \frac{4}{9}(9 - x^2)$$

$$y = \pm \frac{2}{3} \sqrt{9 - x^2}$$

[1]

$$A = 4 \int_0^3 \frac{2}{3} \sqrt{9 - x^2} \, dx$$

$$A = \frac{8}{3} \int_0^3 \sqrt{9 - x^2} \, dx$$

[1]

$$\text{CR} \quad x = \pm \frac{3}{2} \sqrt{4 - y^2}$$

$$A = 6 \int_0^2 \sqrt{4 - y^2} \, dy$$

$$(b) \quad A = \frac{8}{3} \times \frac{\pi}{4} \times 3^2$$

$$A = 6\pi \text{ units}^2 \quad (2)$$

Question 4

$$(a) \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

$$\frac{2x}{a^2} + \frac{2y}{b^2} \cdot \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = -\frac{b^2 x}{a^2 y}$$

$$\text{at } P, m_r = \frac{-b \cos \theta}{a \sin \theta}$$

$$\therefore m_n = \frac{a \sin \theta}{b \cos \theta} \quad [2]$$

$$\Rightarrow y - b \sin \theta = \frac{a \sin \theta}{b \cos \theta} (x - a \cos \theta)$$

$$b y \cos \theta - b^2 \sin \theta \cos \theta = a x \sin \theta - a^2 \sin \theta \cos \theta$$

$$a x \sin \theta - b y \cos \theta = (a^2 - b^2) \sin \theta \cos \theta$$

[1]

3

$$(b) (y=0) \quad a x \sin \theta = (a^2 - b^2) \sin \theta \cos \theta$$

$$x_a = \frac{a^2 - b^2}{a} \cos \theta \quad [1]$$

$$\frac{OQ}{ON} = \frac{\frac{a^2 - b^2}{a} \cos \theta}{a \cos \theta}$$

$$= \frac{a^2 - b^2}{a^2}$$

$$= 1 - \frac{b^2}{a^2}$$

$$= e^2$$

[1]

2

5

Question 5

$$(a) \frac{x^2 - x + 2}{(x+1)(x^2+1)} = \frac{a}{x+1} + \frac{bx+c}{x^2+1}$$

$$x^2 - x + 2 = a(x^2+1) + (bx+c)(x+1)$$

$$(x=-1): 4 = 2a$$

$$\underline{a=2} \quad [1]$$

$$(x=0): 2 = a+c$$

$$\underline{c=0} \quad [1]$$

$$(x=1): 2 = 2a + 2(b+c)$$

$$2 = 4 + 2b$$

$$\underline{b=-1} \quad [1]$$

$$\therefore \frac{x^2 - x + 2}{(x+1)(x^2+1)} = \frac{2}{x+1} - \frac{x}{x^2+1}$$

3

$$(b) \int \frac{x^2 - x + 2}{(x+1)(x^2+1)} dx = \int \left(\frac{2}{x+1} - \frac{x}{x^2+1} \right) dx$$

$$= 2 \ln(x+1) - \frac{1}{2} \ln(x^2+1) + C \quad [2]$$

$$= \ln \left[\frac{(x+1)^2}{\sqrt{x^2+1}} \right] + C \quad [1]$$

3

• Question 6

Prove $1^2 + (1^2 + 2^2) + \dots + \sum_{k=1}^n k^2 = \frac{n}{12} (n+1)^2 (n+2)$

Test $n=1$: LHS = $1^2 = 1$ RHS = $\frac{1}{12} (2)^2 (3)$
 $= 1$

$\therefore$ true for $n=1$ [1]

Assume true for $n = \text{GEEK}$

~~Prove~~
 Prove true for $n+1$

$$\text{LHS} = 1^2 + (1^2 + 2^2) + \dots + \sum_{k=1}^n k^2 + \sum_{k=1}^{n+1} k^2 \quad [1]$$

$$= \frac{n}{12} (n+1)^2 (n+2) + \sum_{k=1}^{n+1} k^2 \quad (\text{by assumption})$$

$$= \frac{n}{12} (n+1)^2 (n+2) + \frac{1}{6} (n+1) (n+2) (2n+3) \quad [1]$$

$$= \frac{1}{12} (n+1) (n+2) [n(n+1) + 2(2n+3)]$$

$$= \frac{1}{12} (n+1) (n+2) (n^2 + 5n + 6)$$

$$= \frac{1}{12} (n+1) (n+2) (n+2) (n+3)$$

$$= \frac{n+1}{12} (n+2)^2 (n+3) \quad [1]$$

$\therefore$ BLAH BLAH BLAH

4

Question 7

$$(a) \quad \frac{x^2}{a^2} - \frac{(mx+k)^2}{b^2} = 1$$

$$b^2 x^2 - a^2 (mx+k)^2 = a^2 b^2$$

$$b^2 x^2 - a^2 m^2 x^2 - 2a^2 mkx - a^2 k^2 = a^2 b^2$$

$$(b^2 - a^2 m^2)x^2 - 2a^2 mkx - a^2(b^2 + k^2) = 0 \quad [1]$$

$$\Delta = 0$$

$$4a^4 m^2 k^2 + 4a^2(b^2 - a^2 m^2)(b^2 + k^2) = 0 \quad [1]$$

$$(\div 4a^2) \quad a^2 m^2 k^2 + b^4 + b^2 k^2 - a^2 b^2 m^2 - a^2 m^2 k^2 = 0$$

$$b^2(b^2 - a^2 m^2 + k^2) = 0$$

$$b \neq 0, \therefore k^2 = a^2 m^2 - b^2 \quad [1] \quad (3)$$

$$(b) \quad y = mx + k, (1, 3) \Rightarrow 3 = m + k \quad [1] \Rightarrow k = 3 - m$$

$$a^2 = 4, b^2 = 15 \Rightarrow 4m^2 - 15 = k^2 \quad [2]$$

$$4m^2 - 15 = (3 - m)^2$$

$$4m^2 - 15 = 9 - 6m + m^2$$

$$3m^2 + 6m - 24 = 0$$

$$m^2 + 2m - 8 = 0$$

$$(m + 4)(m - 2) = 0$$

$$m = -4, 2$$

$$k = 7, 1$$

[1]

$$\therefore y = -4x + 7 \text{ \& } y = 2x + 1$$